AMA515A Forecasting and applied time series analysis

Chapter 3 : Box-Jenkins (ARIMA) models

3.1 Stationarity

If a discrete time stochastic process $\{X_t\}$ satisfy the following conditions:

(i) $E(X_t) = \mu$ (constant) for all t
$Var(X_t) = \sigma^2$ (constant) for all t

(ii) $Cov(X_t, X_s) = \Gamma_{t-s}$ (depend on t-s only !)

Then $\{X_t\}$ possesses second order stationarity.

In our following discussion we concentrate on process of this type.

auto covariance function

or $Cov(X_t, X_{t+k}) = \Gamma_k$
$Cov(X_t, X_{t-k}) = \Gamma_k$

does not depend on t.

gamma

3.2 Autocorrelation function of a stationary stochastic process

Recall for a stochastic process which is stationary, the autocovariance function is equal to

$\gamma(k) = E(X_t X_{t+k}) - E(X_t)E(X_{t+k})$ and $\gamma(0) = E(X_t^2) - E^2(X_t) = var(X_t)$.

$= E(X_t^2) - [E(X_t)]^2$

Note: $\gamma(k)$ is known as the autocovariance function.

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We then define the autocorrelation function $\rho(k) = \frac{\gamma(k)}{\gamma(0)}$, k=0, ±1 , ±2, …

3.3 Some simple stochastic models and their autocorrelation function

3.3.1 White noise

independent and identically distributed

$X_t = Z_t$, $t = 0, \pm1, \pm2, \ldots$, where $\{Z_t\}$ is a sequence of iid rv's with zero mean and constant variance.

$E(Z_t) = 0$
$V(Z_t) = \sigma^2$

Then $\gamma(k) = E(X_t X_{t-k}) = \begin{cases} \sigma^2, k = 0 \\ 0, o.w. \end{cases}$ otherwise

and $\rho(k) = \begin{cases} 1, k = 0 \\ 0, k \neq 0 \end{cases}$

3.3.2 First order Autoregressive Process

AR(1)

$X_t = \alpha X_{t-1} + Z_t$, where $|\alpha| < 1$ (3.1)

In general, we have AR(p), that is p^{th} order Autoregressive Model.

and $\{Z_t\}$ is white noise; Z_t, X_{t-k} are uncorrelated for $k \neq 0$. Then assume $\{X_t\}$ is stationary, taking expectation in (3.1),

$E(X_t) = \alpha E(X_{t-1})$

$\Rightarrow \quad \mu = \alpha\mu$

$\Rightarrow \quad \mu = 0$

$E(X_t) = E[\alpha X_{t-1} + Z_t]$
$= \alpha E(X_{t-1}) + E(Z_t)$
$= \alpha E(X_{t-1}) + 0$

and $V(X_t) = \alpha^2 V(X_{t-1}) + V(Z_t) + 2\alpha \mathrm{Cov}(X_{t-1}, Z_t)$

$\Rightarrow \quad V(X_t) = \alpha^2 V(X_t) + \sigma^2$

$V(X_t) - \alpha^2 V(X_t) = \sigma^2$

$\Rightarrow \quad V(X_t) = \sigma^2/(1-\alpha^2)$

$V(X_t)[1-\alpha^2] = \sigma^2$

$V(X_t) = V[\alpha X_{t-1} + Z_t]$
$= \alpha^2 V(X_{t-1}) + V(Z_t)$
$+ 2\alpha \mathrm{Cov}(X_{t-1}, Z_t)$
0

For autocorrelation,

$$E(X_t X_{t-k}) = \alpha E(X_{t-1} X_{t-k}) + E(Z_t X_{t-k})$$
$$\Rightarrow \rho(k) = \alpha\rho(k-1)$$
$$\Rightarrow \rho(k) = \alpha^k, \; as \; \rho(0) = 1$$

phi ϕ

Thm 3.1 : For an AR(1) process, $X_t = \phi X_{t-1} + \varepsilon_t$, $|\phi| < 1$ is a necessary and sufficient condition for stationarity

Proof: By repeated substitution from the model we get

$$X_t = \phi(\phi X_{t-2} + \varepsilon_{t-1}) + \varepsilon_t = \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j} . \qquad (3.2)$$

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We then make use of the theory of linear difference equation. See for example, Wei, 1993.

We first solve the homogenous equation $X_t - \phi X_{t-1} = 0$ to give the complementary function $X_t = k\phi^t$ where k depends on the initial condition. The infinite moving average representation (3.2) is a particular solution to this difference equation. The general solution to the original AR(1) difference equation is then the sum of the complementary function and the particular solution, i.e.

$$X_t = k\phi^t + \sum_{j=0}^{\infty} \phi^j \varepsilon_{t-j} \qquad (3.3)$$

When $|\phi| < 1$ the complementary function tends to zero for large t, so only the second term of (3.3), the particular solution, remains and the mean of X_t is independent of t.

$E(X_t)=0$. In addition the variance of the second term of (3.3) converges for $|\phi|<1$, so the model is second order stationary. (Exercise : show that $E(X_tX_{t-k})=\gamma(k)=\frac{\sigma^2\phi^k}{1-\phi^2}$.)

Conversely from (3.2) it is clear that the variance of X_t is

$$\sigma^2\sum_{j=0}^{\infty}\phi^{2j}\text{, (assume variance of } \varepsilon_t \text{ is } \sigma^2\text{)}, = \frac{\sigma^2}{1-\phi^2}$$

and for this sum to be finite we need $|\phi|<1$.

MA(q)

3.3.3 Finite order moving average process

The process is defined by $X_t=\sum_{i=0}^{m}\beta_i Z_{t-i}$, where $\{Z_t\}$ is white noise and conventionally we set $\beta_0=1$.

MA(m)

Clearly $E(X_t)=0$, $Var(X_t)=\sigma^2\sum_{i=0}^{m}\beta_i^2$ and

$$\gamma(k)= Cov(X_t,X_{t+k})=E(X_tX_{t+k})=E[\sum_{i=0}^{m}\beta_iZ_{t-i}\sum_{j=0}^{m}\beta_jZ_{t-j+k}]$$

$$=\sum_{i=0}^{m}\sum_{j=0}^{m}\beta_i\beta_jE(Z_{t-i}Z_{t-j+k}) \qquad E(Z_{t-i}Z_{t-j+k})=\sigma^2$$

$$=\begin{cases}\sigma^2\sum_{i=0}^{m-k}\beta_i\beta_{i+k}, & k=0,1,\ldots,m\\ 0, & k>m\end{cases} \qquad (*)$$

$i=j-k$

$j=i+k$ if this happens

This is because $E(Z_{t-i}Z_{t-j+k})\neq 0$ only when j=i+k.

So $\rho(k)=1$ if k=0 and

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$$\rho(k)=\begin{cases}\sum_{i=0}^{m-k}\beta_i\beta_{i+k}\Big/\sum_{i=0}^{m}\beta_i^2, & k=1,2,\ldots,m\\ 0, & k>m\end{cases}$$

$$\rho(k)=\frac{\gamma(k)}{\gamma(0)}=\frac{\sum_{i=0}^{m-k}\beta_i\beta_{i+k}}{\sum_{i=0}^{m}\beta_i^2}$$

i.e, a cut-off at lag m.

NB1. The following diagram is instructive

i	j
$1.Z_t$	$1.Z_{t+k}$
$\beta_1 Z_{t-1}$	$\beta_1 Z_{t-1+k}$
⋮	⋮
$\beta_i Z_{t-i}$	⋮
⋮	⋮
⋮	$\beta_{i+k} Z_{t-j+k}$
⋮	⋮
$\beta_m Z_{t-m}$	$\beta_m Z_{t-m+k}$

NB2. In (*), to evaluate $\sum_{i=0}^{m}\sum_{j=0}^{m}\beta_i\beta_j E(Z_{t-i}Z_{t-j+k})$, from the above table, we see that it will be simplified to the form $\sum_{i=0}^{m-k}\beta_i\beta_{i+k}E(Z_{t-i}Z_{t-(i+k)+k}) = \sum_{i=0}^{m-k}\beta_i\beta_{i+k}\sigma^2$. Because

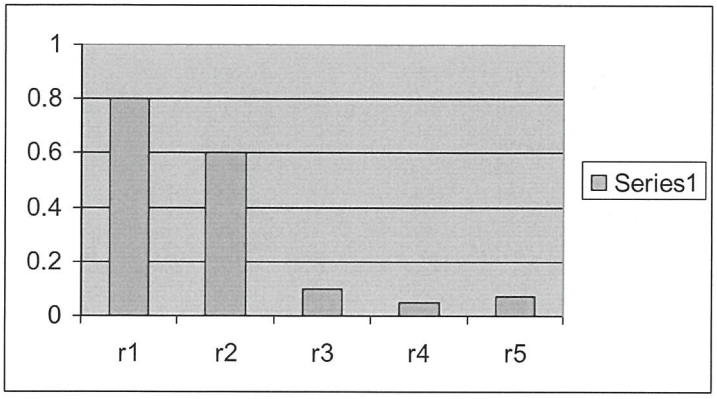
t-i = t-j+k implies j= i+k.

NB3. i+k is at most m, thus i is at most m-k, so we have $\sigma^2\sum_{i=0}^{m-k}\beta_i\beta_{i+k}, k=0,1,...,m$.

Moreover, j=i+k and both i and j are at most m, thus k is at most m in (*) for the existence of non-zero terms.

Example: For MA(2), we have

$$\rho(1)=\frac{\beta_1+\beta_1\beta_2}{1+\beta_1^2+\beta_2^2},\quad \rho(2)=\frac{\beta_2}{1+\beta_1^2+\beta_2^2}.$$



The above diagram is the theoretical correlogram of an MA(2) process.

Example: For an invertible MA(2) process

$$y_t = \varepsilon_t - 0.2\varepsilon_{t-1} + \theta_2\varepsilon_{t-2},$$

$\theta_1 = -0.2$

you are given that $\rho_1 = -0.1$. Determine ρ_2.

Solution: $\rho(1)$ $\rho(2)$

$$-0.1 = \rho_1 = \frac{\theta_1 + \theta_1\theta_2}{1 + \theta_1^2 + \theta_2^2} \Rightarrow -0.1 = \frac{-0.2 - 0.2\theta_2}{1 + 0.04 + \theta_2^2}$$

turn the condition upside down, or inside out or reversible in order

Solving the last equation, we get θ_2 =-0.4 or 2.4. To satisfy the invertibility condition,

θ_2 =-0.4. Then $\rho_2 = -1/3$.

⇑ for MA models

equivalent to say that

the model is stationary.

for AR models.

3.3.4 Random Walk

Consider $X_t = X_{t-1} + Z_t$, where $\{Z_t\}$ is white noise, Z_t and X_{t-k} are uncorrelated for k>0. Then it is easy to see X_t is mean stationary, but

$$\text{Var}(X_t) = \text{Var}(X_{t-1}) + \sigma^2$$

$$= \text{Var}(X_{t-2}) + 2\sigma^2$$

$= \ldots$

$= \mathrm{Var}(X_0) + t\sigma^2$

Therefore the process is not stationary. However,

$Y_t = \nabla X_t = X_t - X_{t-1} = Z_t$ is stationary.

3.4 A general class of linear process

To extend our discussion, it is convenient to introduce 2 operators,

(i) Backward shift $BX_t = X_{t-1}$

(ii) Differecnce $\nabla X_t = X_t - X_{t-1}$ (Backward difference operator) ∇

These operators obey simple algebraic rules, e.g. $B^2 X_t = B(BX_t) = B(X_{t-1}) = X_{t-2}$.

Also $\nabla \equiv 1-B$,

so $\nabla^2 = (1-B)^2 = 1-2B+B^2$, which implies $\nabla^2 X_t = X_t - 2X_{t-1} + X_{t-2}$.

This agrees with $\nabla^2 X_t = \nabla(\nabla X_t) = \nabla(X_t - X_{t-1})$

$= X_t - X_{t-1} - (X_{t-1} - X_{t-2})$

$= X_t - 2X_{t-1} + X_{t-2}$

$X_t - \alpha X_{t-1} = Z_t$

$(1-\alpha B) X_t = Z_t$

Also expression like $(1-\alpha B)^{-1} X_t$ can be regarded as $(1 + \alpha B + \alpha^2 B^2 + \ldots) X_t$.

$= \frac{1}{1-\alpha B} X_t$

Examples:

(i) $X_t = \alpha X_{t-1} + Z_t$ is eq. to $(1 - \alpha B) X_t = Z_t$ AR(1)

(ii) $X_t = \sum_{i=0}^{m} \beta_i Z_{t-i}$ is equal to $X_t = \Theta(B) Z_t$, where $\Theta(B)$ is a polynomial in B $\Theta(B) = \sum_{i=0}^{m} \beta_i B^i$, $\beta_0 = 1$ MA(m)

(iii) $X_t = X_{t-1} + Z_t$ is equal to $\nabla X_t = Z_t$. AR(1)

3.4.1 Infinite and finite order moving average process

Infinite order

Let Z_t be white noise. Consider $X_t = \alpha X_{t-1} + Z_t$

$X_t - \alpha X_{t-1} = Z_t$

$(1 - \alpha B) X_t = Z_t$

Write $(1-\alpha B)X_t = Z_t$, which suggests $X_t = (1-\alpha B)^{-1} Z_t = \sum_{i=0}^{\infty} \alpha^i B^i Z_t = \sum_{i=0}^{\infty} \alpha^i Z_{t-i}$.

Check:

$X_t = \alpha X_{t-1} + Z_t = \alpha(\alpha X_{t-2} + Z_{t-1}) + Z_t$

$= \alpha^2 X_{t-2} + \alpha Z_{t-1} + Z_t$

$= \alpha^3 X_{t-3} + \alpha^2 Z_{t-2} + \alpha Z_{t-1} + Z_t$

$= \ldots = \alpha^t X_0 + \alpha^{t-1} Z_1 + \alpha^{t-2} Z_2 + \cdots + \alpha^2 Z_{t-2} + \alpha Z_{t-1} + Z_t$

This shows that an AR(1) process can be represented as an infinite order moving average process.

AR(1) ≡ MA(∞)

Finite order

$\beta_0 = 1$

$X_t = \Theta(B) Z_t = (\beta_0 + \beta_1 B + \beta_2 B^2 + \ldots + \beta_m B^m) Z_t = \sum_{i=0}^{m} \beta_i Z_{t-i}$ MA(m)

An important concept related to moving average (MA) process is invertibility.

For an MA(1) process,

$X_t = Z_t + \beta Z_{t-1}$, $|\beta| < 1$ (3.4)

It is easy to see that $\rho(0)=1$, $\rho(1)=\beta/(1+\beta^2)$, $\rho(k)=0$ for $k>1$.

Now consider the following process,

$X_t = Z_t + \frac{1}{\beta} Z_{t-1} = Z_t + \frac{B}{\beta} Z_t = (1 + \frac{B}{\beta}) Z_t$ (3.5) $|\frac{1}{\beta}| > 1$

$|\beta| < 1$

Then clearly it has the same ACF(autocorrelation function) as the previous process.

So there is an indeterminacy problem. not precise, uncertain

And for (3.4), observe $X_t = (1+\beta B) Z_t$

$X_t = Z_t + \beta Z_{t-1}$

$= (1+\beta B) Z_t$

$\Rightarrow Z_t = (1+\beta B)^{-1} X_t$

$= X_t - \beta X_{t-1} + \beta^2 X_{t-2} - \ldots,$

$(1+\beta B)^{-1} = 1 - \beta B + \beta^2 B^2 - \beta^3 B^3 + \beta^4 B^4 - \cdots$

Then $Z_t = (1 - \beta B + \beta^2 B^2 - \beta^3 B^3 + \beta^4 B^4 - \ldots) X_t$

$= X_t - \beta X_{t-1} + \beta^2 X_{t-2} - \beta^3 X_{t-3} + \beta^4 X_{t-4} - \cdots$

whence the coefficients on the RHS is decaying and convergent.

For (3.5),

$$Z_t = X_t - \frac{1}{\beta} X_{t-1} + \frac{1}{\beta^2} X_{t-2} - \ldots, \quad (3.5^*)$$

The coefficients on the RHS are not decaying and not convergent. The representation is not meaningful as that suggests the error's dependence on X_t increases as we go back in time. In general, for MA process $X_t = \Theta(B) Z_t$, if the representation $Z_t = \Theta(B)^{-1} X_t$ has this behavior, then the time series X_t is not invertible. Intuitively, a non-invertible time series may have problem in statistical estimation.

For the MA(1), restricting $|\beta|<1$ will resolve both the indeterminacy and convergency problem. For general MA(q) process, we have the following theorem.

Thm.: A necessary and sufficient condition for a finite order MA process to be invertible is that all the roots of the polynomial $\Theta(B)$ lie outside the unit circle.

3.4.2 Finite order autoregressive processes

Consider $X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + \ldots + \alpha_m X_{t-m} + Z_t$. (3.6)

In operator form, $\Phi(B) X_t = Z_t$, where $\Phi(.)$ is a polynomial of degree m.

Temporarily assume stationarity, then

$E(X_t)=0$, so multiply (3.6) by X_{t-k} and take expectation gives

$$\gamma(k) = \alpha_1 \gamma(k-1) + \alpha_2 \gamma(k-2) + \ldots + \alpha_m \gamma(k-m), \; k = 1,2,\ldots.$$

In particular,

$$\rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2) + \ldots + \alpha_m \rho(k-m), \; k = 1,2,\ldots \quad (3.7)$$

(3.7) are known as Yule-Walker equations. In principle, they can be solved to give $\rho(k)$, in terms of given α_I, and vice versa.

Write (3.7) as

$$\begin{pmatrix} \rho(1) \\ \cdot \\ \cdot \\ \cdot \\ \rho(m) \end{pmatrix} = \begin{pmatrix} 1 & \rho(1) & \dots & \rho(m-1) \\ \rho(1) & 1 & \dots & \rho(m-2) \\ \vdots & & & \\ \rho(m-1) & \dots & & 1 \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \cdot \\ \cdot \\ \cdot \\ \alpha_m \end{pmatrix}.$$

Because $\rho(k) = \rho(-k)$ for stationary $\{X_t\}$, i.e, $\rho = \mathbf{R}\alpha$

and provided R^{-1} is defined, we have $\alpha = \mathbf{R}^{-1}\rho$, i.e, given the 1st m autocorrelations, or estimates of the corr. parameters, the mth order autoregresssive process may be obtained, or estimated.

Thm. A necessary and sufficient condition for the stationarity of the autoregressive process $\Phi(B)X_t = Z_t$, where $\Phi(B)$ is a polynomial in B of order m, is that the roots of the equation

$$\Phi(B) = 0$$

should all lie outside the unit circle.

Proof: We first show the AR(2) case, i.e., $X_t = \phi_1 X_{t-1} + \phi_2 X_{t-2} + \varepsilon_t$

The solution to the homogenous difference equation $X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} = 0$ depends on the roots r_1, r_2 of the equation $1 - \phi_1 z - \phi_2 z^2 = 0$ (See Wei, 1993) which we shall assume are distinct although they may be complex. The complementary function is of the form $X_t = k_1(1/r_1)^t + k_2(1/r_2)^t$, where k_1 and k_2 are constants depending on initial conditions. The particular solution is the infinite MA representation $X_t = \phi^{-1}(B)\varepsilon_t = \psi(B)\varepsilon_t$. The general solution to the AR difference equation is therefore $k_1(\frac{1}{r_1})^t + k_2(\frac{1}{r_2})^t + \sum_{j=0}^{\infty}\psi_j\varepsilon_{t-j}$.

(please see p. 2 , (3.3))

For the model to have a constant mean, the complementary function must decay for large t, so we require $|1/r_1| < 1$ and $|1/r_2| < 1$. Moreover, the AR polynomial takes the form

$$\phi(B) = \left(1 - \frac{1}{r_1}B\right)\left(1 - \frac{1}{r_2}B\right),$$

where r_1, r_2 are its roots. A particular solution is $X_t = \psi(B)\varepsilon_t = \phi^{-1}(B)\varepsilon_t$, so using partial fractions we can write

$$\phi^{-1}(B) = \frac{a}{\left(1-\frac{1}{r_1}B\right)} + \frac{b}{\left(1-\frac{1}{r_2}B\right)}$$

where a and b are constants. Each of the terms in $\phi^{-1}(B)$ can be expanded - for instance, the first is

$$a\left[1+\frac{1}{r_1}B+\left(\frac{1}{r_1}B\right)^2+\ldots\right], \qquad b\left[1+\frac{1}{r_2}B+\left(\frac{1}{r_2}B\right)^2+\ldots\right]$$

so if the variance of $X_t = \psi(B)\varepsilon_t = \phi^{-1}(B)\varepsilon_t$ is to converge we require that

$$\frac{1}{|r_1|}<1 \text{ and } \frac{1}{|r_2|}<1. \Rightarrow |r_1|>1 \text{ and } |r_2|>1$$

Hence the condition for an AR(2) model to be stationary is that both the roots r_1 and r_2 of the AR polynomial are outside the unit circle.

Similar logic can be applied to the general AR(p) model. Assuming that the roots of the corresponding AR polynomial $\phi(z) = 1-\phi_1 z-\ldots-\phi_p z^p$ are dintinct and denoting these $r_1, \ldots r_p$ the complementary function is

$$k_1\left(\frac{1}{r_1}\right)^t + k_2\left(\frac{1}{r_2}\right)^t + \ldots + k_p\left(\frac{1}{r_p}\right)^t.$$

Thus similar to the AR(2) argument if the roots are all outside the unit circle the complementary function will decay to zero for large t and the model will be stationary.

Conversely, i.e. , the model is stationary and thus has finite variance. Let $\lambda_i = 1/r_i$, then $X_t = \phi^{-1}(B)\varepsilon_t = \left(\prod_{i=1}^{p}(1-\lambda_i B)\right)^{-1}\varepsilon_t$. Assume again the roots are distinct, then

$X_t = \left\{\sum_{i=1}^{p}\frac{a_i}{1-\lambda_i B}\right\}\varepsilon_t$, for suitable a_i's. Therefore,

$$X_t = \sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i(\lambda_i B)^j \varepsilon_t = \sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i \lambda_i^j B^j \varepsilon_t$$

$$= \sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i \lambda_i^j \varepsilon_{t-j}$$

$a_i \frac{1}{1-\lambda_i B}$

$$X_t = \sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i \lambda_i^{\,j} \varepsilon_{t-j}$$

So $\mathrm{Var}(X_t)=\sigma^2\sum_{i=1}^{p}\sum_{j=0}^{\infty} a_i^{\,2}\lambda_i^{\,2j} = \sigma^2\sum_{i=1}^{p} a_i^{\,2}\sum_{j=0}^{\infty}\lambda_i^{\,2j}$. $\mathrm{Var}(X_t)$ is finite implies the absolute values of all λ_i's are less than one.

Example: An AR(2) process

Let $X_t= 0.7X_{t-1}-0.1X_{t-2}+Z_t$, $\{Z_t\} \sim NID(0,\sigma^2)$.

Then $(1-0.7B+0.1B^2)X_t=Z_t$. The roots of the backshift polynomial in B, i.e.,

$1-0.7B+0.1B^2=0$, are 2 and 5. Thus the AR(2) process is stationary.

3.4.3 Mixed processes(ARMA processes)

The equation $\alpha_0 X_t - \alpha_1 X_{t-1} - \ldots - \alpha_p X_{t-p} = Z_t + \beta_1 Z_{t-1} + \ldots + \beta_q Z_{t-q}$

Where $\{Z_t\}$ is white noise and conventionally $\alpha_0=1$ defines a mixed autoregressive moving average process of order (p,q), ARMA(p,q).

In operator notation, we have $\Phi(B)X_t = \Theta(B)Z_t$.

Autocorrelation functions are available in principle via the equation

$$X_t = \frac{\Theta(B)}{\Phi(B)} Z_t, \quad \Rightarrow \quad X_t = W(B)Z_t$$

where $W(B) = \frac{\Theta(B)}{\Phi(B)}$

and $\Phi^{-1}(B)$ is an infinite series, $\therefore$ MA(∞)

which expresses X_t as a MA(∞) process.

Example: An ARMA(2,1) process

Consider the process $X_t - 0.75X_{t-1} + 0.5625X_{t-2} = Z_t + 1.25Z_{t-1}$, $\{Z_t\} \sim NID(0,\sigma^2)$

The AR polynomial $\phi(B) = 1-0.75B+0.5625B^2$ has zeros at $B=2(1\pm i\sqrt{3})/3$, which lie outside the unit circle. The process is stationary. On the other hand, the MA polynomial $\theta(B)$ has a zero at b=-0.8 and hence the process is not invertible.

Parsimony

many

In practice, may sets of data can reasonably be modelled by either a high order MA process or a lower order AR process. If so, we positively refer the version with fewer parameters. Similarly, an ARMA(p,q) process of lower order can often be made to fit data as either high order AR or MA process.

3.4.4 Low order models

Experience suggest that ARMA(p,q) process with p+q ≤ 2 constitute an extremely useful class of parsimonious models for stationary time series data. We can summarize main features of AR(2), MA(2), and ARMA(1,1) processes.

not generous

$\frac{-b \pm \sqrt{b^2-4ac}}{2a}$

AR(2)

$X_t = \alpha_1 X_{t-1} + \alpha_2 X_{t-2} + Z_t \Rightarrow X_t - \alpha_1 X_{t-1} - \alpha_2 X_{t-2} = Z_t$

$\Rightarrow \Phi(B)X_t = Z_t, \Phi(B) = 1 - \alpha_1 B - \alpha_2 B^2$ $\quad (1 - \alpha_1 B - \alpha_2 B^2) X_t = Z_t$

and $\Phi(B) = 0 \Leftrightarrow B = (\alpha_1 \pm \sqrt{\alpha_1^2 + 4\alpha_2})/(-2\alpha_2)$. $\quad B = \frac{\alpha_1 \pm \sqrt{\alpha_1^2 + 4\alpha_2}}{-2\alpha_2}$

For stationarity, require $|B| > 1$.

The Yule-Walker equations become

P.8 (3.7) with m = 2

$$\rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2), k \geq 0 \qquad (3.8)$$

In principle, can solve recursively from initial values $\rho(0) = 1, \rho(1) = \frac{\alpha_1}{1-\alpha_2}$.

Thus $\rho(2) = \frac{\alpha_1^2}{1-\alpha_2} + \alpha_2$, …etc.

imply

Note that constraints on α_1, α_2 implie constraints on $\rho(1)$ & $\rho(2)$. Alternatively, since (3.8) is a linear , 2nd order difference equation, it can be solved explicitly. In fact, for (3.8), i.e., $\rho(k) = \alpha_1 \rho(k-1) + \alpha_2 \rho(k-2), k \geq 0$, the general solution is

See P. 9

$$\rho(k) = A_1 \lambda_1^k + A_2 \lambda_2^k, \text{ where } \lambda_1, \lambda_2 \text{ are roots of } \lambda^2 - \alpha_1 \lambda - \alpha_2 = 0.$$

Values of A_1, A_2 can be obtained from using the initial values of $\rho(0)$ & $\rho(1)$.

Three distinct forms of solution:

(i) $\alpha_1^2 + 4\alpha_2 > 0 \Rightarrow$ mixtures of two exponentially decaying function for ρ(k)

(ii) $\alpha_1^2 + 4\alpha_2 < 0 \Rightarrow \lambda = ae^{\pm i\theta}$, ρ(k) is a damped harmonic. (ρ will be repeated with the same pattern)

(iii) $\alpha_1^2 + 4\alpha_2 = 0 \Rightarrow \rho(k) = (A_1 + A_2 k)\lambda^k$, an exponential decay.

then $\lambda_1 = \lambda_2 = \lambda$

MA(2) Process → $(1+\beta_1 B+\beta_2 B^2)Z_t$

$X_t = Z_t + \beta_1 Z_{t-1} + \beta_2 Z_{t-2}$, equivalently $X_t = \Theta(B)Z_t$

For invertibility, require roots of $1 + \beta_1 B + \beta_2 B^2 = 0$ to be outside the unit circle, i.e.,

$$\left| \frac{-\beta_1 \pm \sqrt{\beta_1^2 - 4\beta_2}}{2\beta_2} \right| > 1$$

which implies the same restriction on $(-\beta_1,-\beta_2)$ as for (α_1, α_2) in AR(2) process. Again restrictions on $-\beta_1,\beta_2$ implies restrictions on ρ(1) and ρ(2).

ARMA(1,1)

$X_t - \alpha X_{t-1} = Z_t + \beta Z_{t-1}$ => $(1-\alpha B)X_t = (1+\beta B)Z_t$

$\Rightarrow \left\{\frac{1+\beta B}{1-\alpha B}\right\} Z_t$. $\frac{1}{1-\alpha B} = 1+\alpha B+\alpha^2 B^2+\alpha^3 B^3+\cdots = \sum_{i=0}^{\infty} \alpha^i B^i$

Thus $|\alpha|<1$, $|\beta|<1$ are together sufficient to give a stationary , invertible process $\{X_t\}$.

MA(∞) Consider the infinite order moving average representation,

$$X_t = (1+\beta B)\sum_{i=0}^{\infty} \alpha^i B^i Z_t = \left\{1 + \sum_{i=1}^{\infty} \alpha^{i-1}(\alpha+\beta)B^i\right\} Z_t$$

$$= \sum_{i=0}^{\infty} b_i Z_{t-i}\,, \text{ where } b_i \text{ to be determined}$$

By direct calculation, $b_0 = 1$, $b_1 = \alpha + \beta$, ..., $b_i = \alpha^{i-1}(\beta+\alpha)$, *for* $i \geq 1$

then using results of section 3.3.3,

$$\rho(k) = (\sum_{i=0}^{\infty} b_i b_{i+k}) / \sum_{i=0}^{\infty} b_i^2$$

and $$\sum b_i^2 = 1 + (\alpha+\beta)^2 \sum_{i=1}^{\infty} \alpha^{2i-2} = 1 + \frac{(\alpha+\beta)^2}{1-\alpha^2},$$

$$\sum_{i=0}^{\infty} b_i b_{i+k} = \alpha^{k-1}(\alpha+\beta) + \frac{(\alpha+\beta)^2 \alpha^k}{1-\alpha^2}$$

and $\rho(k) = [\alpha^{k-1}(\alpha+\beta)(1+\alpha\beta)]/(1+2\alpha\beta+\beta^2)$, for $k \geq 1$

i.e., exponential decay, $\rho(k) = \alpha\rho(k-1)$, for $k \geq 2$.

Example: For an ARMA(2,1) process

$$y_t = 0.3y_{t-1} + 0.2y_{t-2} + \varepsilon_t + \theta\varepsilon_{t-1}$$

You are given $\rho_1 = 0.8, \rho_2 = 0.44$. Find ρ_4.

Solution: Use the Yule-Walker equations.

$$\rho_4 = 0.3\rho_3 + 0.2\rho_2 = 0.3\rho_3 + 0.088$$
$$\& \quad \rho_3 = 0.3\rho_2 + 0.2\rho_1 = 0.292$$

hence $\rho_4 = 0.1756$.

3.5 Integrated process

Recall the Markov process $X_t = X_{t-1} + Z_t$, which is nonstationary but can be transformed to a stationary process by differencing. More generally, consider the process

$$\Omega(B)X_t = \Theta(B)Z_t \qquad (3.9)$$

where $\{Z_t\}$ is white noise, and $\Omega(B) = \Phi(B)(1-B)^d$.

$\Phi(B)$, $\Theta(B)$ are polynomials of degree p,q respectively, all of their roots outside unit circle. Note that , so we can write (3.9) as

d ← is the seasonal ~~factor~~ operator

p ↓ q

$\Phi(B)\{\nabla^d X_t\} = \Theta(B)Z_t \Leftrightarrow \Phi(B)(1-B)^d X_t = \Theta(B)Z_t$

i.e., the dth difference of $\{X_t\}$ is ARMA(p,q), we call (3.9) an autoregressive integrated moving average process or $\{X_t\}$ is an ARIMA(p,d,q) process. $\{X_t\}$ is nonstationary, so $\rho(k)$ does not exist. However, $\hat{\rho}(k)$ can be useful in identifying ARIMA process. Typically slow decay from $\hat{\rho}(1) \approx 1$ suggest an integrated process a plausible model. → possible

s

Warning : over difference an already stationary series will often produce an undesirable increase in variability.

3.6 Seasonal models

seasonally { Spring, Summer, Autumn, Winter } or Monthly

$1-B^s$ is the seasonal operator

The seasonal difference operator $\nabla_s X_t = X_t - X_{t-s} = (1-B^s)X_t$ can be used to eliminate seasonal nonstationarity. Formally, consider seasonal ARIMA models of the form $\Phi(B)(1-B)^{d_1}(1-B^s)^{d_2}X_t = \Theta(B)Z_t$.

p d_1 d_2 q

$\Phi(B)(1-B)^d X_t = \Theta(B)Z_t$

p d q

∴ ARIMA(p, d, q)

$\Phi(B)(1-B)^{d_1}(1-B^s)^{d_2}X_t = \Theta(B)Z_t$

$ARIMA(p, d_1, q) \times (0, d_2, 0)_s$